Audit of the External Implicit Incidence LP Theorem

Research note

4 October 2026

Abstract

Theorem 5.3 of the directed-reachability note [1] is not established by the supplied proof. The main unresolved issue is the stability of the compiled, rounded Lee–Sidford execution. A relative linear-solve perturbation claim in its proof is false, and the cited Lee–Sidford theorem does not supply the broader interface being used. There are also two smaller implementation omissions, for which we give repairs. This audit does not disprove the claimed query bound. The $\widetilde{O}(n^{3/2})$ terminal-mincut and $\widetilde{O}(n^{7/4})$ global-mincut consequences in our companion note remain conditional.

1 Scope and verdict

The audited object is the 28-page PDF *Directed Reachability from Cut Queries in $\widetilde{O}(n^{3/2})$ Queries*, downloaded on 4 October 2026, with SHA-256

`537ff24b423ecbf41121d5929bcf09b70bbc21de069ab3c6ea89c4a25a54ddb7.`

Page numbers below are the printed page numbers. We compared its Proposition 5.1, Theorem 5.3, and Lemma 6.1 with the actual algorithms and proofs in Lee–Sidford [2, 3]; this is not an inference from their headline iteration bounds.

Component	Audit finding
Section 6 stable execution	Unresolved proof obligation. The cited result does not cover all the approximations and the replacement feasibility repair.
Page 26 solve-error transfer	False as a general perturbation claim; an explicit incidence-plus-star example appears below.
Proposition 5.1 operation list	Omits the minimization defining centrality. A Newton-norm substitute repairs this part with adjusted invariant margins.
Section 5.4 scalar search	The overlapping-interval branch is underspecified. A middle-third update with an incumbent repairs it.
Cut-query and spectral primitives	No blocking defect found in the arguments examined; this is not a machine-checked certification.
Exact star repair	The affine-feasibility identity and its local error-amplification bound are correct.

There is no demonstrated counterexample to the *statement* of Theorem 5.3. There is a counterexample to a step in its proof. Until a replacement stability proof is supplied, the theorem cannot remove the assumption in `directed-mincut-incidence-lp.tex`. Our global-to-terminal reduction, with query cost $\widetilde{O}(nh + (n/h)Q_{st}(n))$, remains independent of that assumption.

2 The principal gap: the imported stability interface

Proposition 5.1, pp. 17–18 of [1], asserts that approximate row reductions, mixed-ball support points, and normal solves can all replace their exact counterparts with fixed inverse-polynomial errors. Lemma 6.1, pp. 25–26, then asserts that the same is true after repairing each direction on the explicit $+I_r$ rows.

The supporting citation does not state this interface:

- Lee–Sidford [2], Theorem 40 and Algorithms 1–3, provide the rank-sensitive path-following analysis, with approximate Lewis weights. The paper explicitly defers the treatment of inexact linear solves to its earlier version.
- Lee–Sidford [3], Section 8 and Theorem 28, analyze a particular stable implementation given an energy-relative normal-system solver. That implementation maintains a normal-force estimate $A\eta$, uses the resulting stable primal step, and periodically reduces infeasibility. Its conclusion allows small affine infeasibility.
- The implicit note instead rounds analytic coordinate reductions, uses an approximate support optimizer, and enforces exact feasibility through a different right inverse. The cited theorem does not by itself prove this modified algorithm stable.

The local estimate in Lemma 6.1 only shows that one repaired direction is close to the intended direction, assuming the polynomial numerical envelope. It does not show that the new iterate preserves the centering and weight-chasing invariants that justify that envelope. Polynomial Lipschitz bounds for individual steps alone are insufficient: composing T maps with Lipschitz constant L can amplify errors by L^T. The required contraction or invariant argument must be made explicit.

A completion should establish, for example, a centrality recurrence of the form

$$\delta_{j+1} \le \left(1 - \frac{1}{Ck_{\mathrm{LS}}}\right)\delta_j + P(MU/\varepsilon)\alpha,$$

together with a compatible bound for the weight-chasing potential, interiority, initialization, the cost switch, and final objective error. Here P must be a fixed polynomial. This would justify a fixed inverse-polynomial precision and the claimed query count.

What the conditioning citations do support. Corollary 68 of [2] bounds the barrier Hessians along the analyzed execution. Lemma 69(1) does bound the normal matrices relative to $A^\top A$; its other parts control successive changes. Combined with the explicit star and bounded row norms, these give useful polynomial spectral bounds. Lemma 67 is a geometric lemma about weighted centers, not itself a path-weight bound; regularization and weight tracking give those bounds. None of these results automatically extends the invariants to the modified rounded execution.

3 A false relative-error transfer claim

On p. 26, the note solves the frozen system

$$\widehat{L} = A^\top \widehat{D} A, \qquad \widehat{b} = A^\top \widehat{D} q,$$

to relative energy error, and claims that its diagonal perturbation bound and polynomial conditioning transfer the same relative guarantee to $L = A^\top D A$, $b = A^\top D q$. Rounding D changes the right-hand side as well as the matrix. Cancellation in b invalidates this inference.

Take one hidden incidence row and two explicit star rows:

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = I_3, \quad q = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad \widehat{D} = \operatorname{diag}(1+\eta, 1, 1), \qquad 0 < \eta < 1.$$

All rows have the required norm bound, and $(1-\eta)D \preceq \widehat{D} \preceq (1+\eta)D$. Nevertheless,

$$L = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad b = 0, \quad \widehat{L} = \begin{pmatrix} 2+\eta & -1-\eta \\ -1-\eta & 2+\eta \end{pmatrix}, \quad \widehat{b} = \eta \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Even an *exact* solve of the frozen system returns

$$\widehat{\pi} = \frac{\eta}{3+2\eta} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \neq 0 = L^{-1}b.$$

Thus no finite multiplicative factor gives $\|\widehat{\pi} - L^{-1}b\|_L \leq \alpha \|L^{-1}b\|_L$. The condition numbers are only 3 and $3+2\eta$. The note's special case for $\widehat{b} = 0$ does not apply. This is a counterexample to the claimed implication from the allowed perturbation bounds, not a claim that every possible rounding rule must choose this particular perturbation.

The issue is not confined to exactly zero right-hand sides. For $q = (1+\tau, -1, 1)^\top$, the relative error of the exact frozen solution is

$$\frac{\|\widehat{\pi} - \pi\|_L}{\|\pi\|_L} = \frac{\eta(3+\tau)}{\tau(3+2\eta)}.$$

It is unbounded as $\tau \downarrow 0$, despite uniformly good matrix conditioning.

A valid replacement estimate. Write $\rho = \|L^{-1/2}(\widehat{L} - L)L^{-1/2}\|_2 < 1$, $\pi = L^{-1}b$, and $\widehat{\pi} = \widehat{L}^{-1}\widehat{b}$. Direct subtraction gives

$$\|\widehat{\pi} - \pi\|_L \leq \frac{\|\widehat{b} - b\|_{L^{-1}} + \rho\|\pi\|_L}{1-\rho}.$$

If $\widetilde{\pi}$ has relative error α for the frozen system, then

$$\|\widetilde{\pi} - \pi\|_L \leq \frac{\|\widehat{b} - b\|_{L^{-1}} + \rho\|\pi\|_L + \alpha\|\widehat{b}\|_{L^{-1}}}{1-\rho}.$$

This is enough for an *absolute* error bound under polynomial envelopes. A stable path-following proof accepting that bound could repair the argument. The current proof does not supply it.

4 An omitted operation, and a repair in exact arithmetic

Lee–Sidford [2], Algorithm 1, explicitly uses

$$\delta_t(x, w) = \min_{\nu \in \mathbb{R}^r} \left\| \frac{tc + W\phi'(x) - A\nu}{w\sqrt{\phi''(x)}} \right\|_{w+\infty}$$

to set the weight-movement ball. This is an optimization over equality multipliers. It is not one of the coordinate reductions in Proposition 5.1, and the implicit note gives no implementation of it. Computing the displayed mixed norm for a specified ν does not perform the minimization.

There is, however, a useful substitute. Let h be the exact projected Newton step, and set

$$d = \|\sqrt{\Phi''(x)}h\|_{w+\infty}.$$

This requires one normal solve and row reductions. Lemmas 10 and 13 of [2] imply, in the path-following regime,

$$\delta \le d \le \left(1 + \frac{1}{16k}\right)\delta, \qquad k = c_k \ge 1.$$

Lemma 1 (Centrality substitution)**.** *Under the hypotheses used in the proof of Theorem 19 of [2], replace the radius* $(1 - 6/(7k))\delta$ *in its weight-chasing ball by* $(1 - 6/(7k))d$*. Assume additionally that* $d \le R$ *and* $\delta \le 1/(320k)$*. With the same exact Newton step and the stated chasing-oracle guarantees, the new centrality is at most* $(1 - 1/(4k))\delta$*.*

Proof. The target-weight motion still fits in the ball because $d \ge \delta$. The ball remains within the required outer radius because $d \le R$. Scaling the ball does not change its inner-to-outer radius ratio, so the other geometric hypotheses of the chasing argument are unchanged. The resulting log-weight motion has mixed norm at most $a\delta$, where

$$\begin{aligned} a &= \left(1 + \frac{1}{2k}\right)\left(1 - \frac{6}{7k}\right)\left(1 + \frac{1}{16k}\right) \\ &= 1 - \frac{33}{112k} - \frac{101}{224k^2} - \frac{3}{112k^3} \le 1 - \frac{33}{112k}. \end{aligned}$$

The Newton step leaves centrality at most $4\delta^2$. Applying Lemma 17 of [2] to the subsequent weight change gives

$$\begin{aligned} \delta_{\text{new}} &\le (1 + 4a\delta)(4\delta^2 + a\delta) \\ &\le \left(1 + \frac{1}{80k}\right)\left(1 - \frac{33}{112k} + \frac{1}{80k}\right)\delta \\ &\le \left(1 - \frac{1}{4k}\right)\delta. \end{aligned}$$

The final inequality uses $33/112 - 1/40 = 151/560 > 1/4$. □

A constant-factor tightening of the outer centrality invariant and path increments accommodates $d \le R$ without changing the asymptotic batch count. This repairs the missing operation in exact arithmetic. With approximate solves and norm evaluations, one still needs a certified upper estimate for d, a tolerance floor, and the stability analysis described above. In particular, positive centrality radii need not have an inverse-polynomial lower bound merely because the path parameters do.

5 The scalar support search needs one more branch

The KKT reduction in Section 5.4 of [1] and its concavity and Lipschitz arguments check out. The description on p. 24 does not specify how the search interval changes when the two trisection value intervals overlap. Returning the better of those two points in this case would be incorrect. Already for one coordinate, $a = \kappa = 1$,

$$G(\tau) = \min\{\tau, 1 - \tau\}, \qquad G(1/3) = G(2/3) = 1/3, \qquad \max G = G(1/2) = 1/2.$$

Equality of the two queried values is not a stopping certificate. If the intended instruction only retains an incumbent, an interval update is still needed.

A repair preserves the claimed logarithmic query count. Maintain a best queried incumbent. With overlapping certified value intervals of width at most ζ, retain the middle third and continue. If the trisection points are $a < b$, then $|G(a) - G(b)| \le 2\zeta$. Concavity bounds the value anywhere in the discarded left third by $G(a)+2\zeta$, and similarly on the right by $G(b)+2\zeta$. Retaining an incumbent prevents these losses from accumulating over iterations. Once the interval width is at most ζ/L_a, the Lipschitz bound finishes an additive $O(\zeta)$ approximation in $O(\log(L_a/\zeta))$ evaluations. This completes the scalar-search logic, but does not establish its tolerance within the full weight-chasing execution.

6 Parts that survive the audit

Oracle operations and spectral construction. The three-query identity for filtered pair incidence is valid. Fixed-bit dyadic row aggregates can be decomposed into bit classes without knowing individual adjacency bits. Public pair predicates can be used in neighbor search. In Section 4, the weighted cluster-spanner argument, the edge-disjoint bundle resistance bound, and independent whole-pair sampling are consistent with constant pair multiplicity. The recursive support masks are public. We found no need in these arguments to inspect an unqueried adjacency bit.

Duplicate rows and leverage estimates. Conceptual identical copies correctly contribute $c_{uv}D_{uv}(e_u - e_v)(e_u - e_v)^\top$, with multiplicity entering linearly. Simultaneous quadratic-form estimates from a sparsifier inverse avoid assigning independent random signs to indistinguishable copies. Conditional on such estimates, the cited approximate Lewis-weight routines use coordinate updates; the initialization homotopy and per-copy lower-bound argument are compatible with the explicit star. Zero explicit rows, permitted by Theorem 5.3 but excluded by Proposition 5.1, should first be removed and optimized separately.

Affine repair. Let $A^\top R_* = I$, let $A^\top h = 0$, and put $h_{\mathrm{rep}} = \widetilde{h} - R_* A^\top \widetilde{h}$. Then

$$A^\top h_{\mathrm{rep}} = 0, \qquad h_{\mathrm{rep}} - h = (I - R_* A^\top)(\widetilde{h} - h).$$

Consequently

$$\|H^{1/2}(h_{\mathrm{rep}} - h)\|_2 \le \big(1 + \|H^{1/2}R_*\|_2 \|A^\top H^{-1/2}\|_2\big)\|H^{1/2}(\widetilde{h} - h)\|_2.$$

For a dyadic raw direction the residual can indeed be aggregated exactly. Thus equality preservation is not the gap. Preserving the box and progress invariants after this repair still needs the missing stability argument.

7 Reproducibility and next proof obligation

The script `implicit_lp_audit_checks.py`, distributed with this note at https://github.com/troyjlee/cut-query-complexity/tree/main/papers/supporting, verifies the cancellation example and its nonzero-right-hand-side variant with exact rational arithmetic, the star-repair identity, the centrality coefficient identity, and the one-dimensional support counterexample. These checks substantiate the displayed examples; they do not certify an interior-point algorithm.

The most useful next step is to prove a finite-precision centering lemma for the actual compiled state, using absolute local-norm errors and the explicit star repair. The result must jointly control centrality, weight chasing, and slack. Alternatively, one could compile the full stable normal-force implementation from [3], while separately accounting for approximate row reductions and support optimization. Neither route is completed by the current external proof.

References

[1] Anonymous research note. *Directed Reachability from Cut Queries in $\widetilde{O}(n^{3/2})$ Queries*. 28-page snapshot, 4 October 2026. https://yangpliu.github.io/repo/directed-reachability-cut-queries/paper.pdf.

[2] Yin Tat Lee and Aaron Sidford. *Solving Linear Programs with $\widetilde{O}(\sqrt{\text{rank}})$ Linear System Solves*. arXiv:1910.08033v2, 31 August 2020. https://arxiv.org/abs/1910.08033v2.

[3] Yin Tat Lee and Aaron Sidford. *Path Finding II: An $\widetilde{O}(m\sqrt{n})$ Algorithm for the Minimum Cost Flow Problem*. arXiv:1312.6713, downloaded 4 October 2026. https://arxiv.org/abs/1312.6713. The downloaded PDF has SHA-256 c7e0690ef9ae7e53390953b65332bd1f1148455e884f0488bafe8beac2e19cdd.